



Engine Operation and Performance Test Witness Report (GT-25)

Department: Airworthiness

Date: 2026-07-20

Head Office: 2nd floor, No. 11, Sixth St., Northern Naft St., Mirdamad St., Tehran, I.R. IRAN

Phone: +98 (21) 29433859 P.O.BOX: 19188-86458

Factory: Malard Road (7th km), Karaj, I.R. IRAN
Phone: +98 (26) 36630010-25 P.O.BOX: 31676-43115

Record No.: AEC_233637

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0 General

0.1 Amendment Record

#	Revision	Amendment Description	Date
1	00	Initial	2026-08-12

0.1 Affected Engines

Type/Model	SN	TC No./TCDS No.	Manufacturer
JT8D-217C, JT8D-219	-	E9NE	Pratt & Whitney (P&W)

0.2 Affected Parts

Nomenclature	OEM P/N	New P/N
LPT Blade 3 rd and 4 th stage	798403 822504	IPA-798403 IPA-822504

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1 Project

1.1 Engine Type/Model

- JT8D-200 series engine
 - JT8D-217C
 - JT8D- 219

1.2 Project No.

- GT-25

2 Certification Specification

Based on Classification Document No. AEC_201337, this project is classified as a non-significant major change. Therefore, the requirements of 14 CFR Part 33, effective December 9, 2022, have been applied.

3 Conformity Checks

3.1 Test Cell Conformity and Capability

The test cell is regularly inspected by the CAA.IRI, and all measurement instrumentation complies with the ISO-17025 standard, ensuring the traceability, accuracy, and reliability of the recorded data.

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In the name of God


 Islamic Republic of Iran
 Civil Aviation Authority

Date: 8 October, 2025
Number: 24120431008

Maintenance Organization Approval Certificate

Reference: IR.145.19

Pursuant to the Civil Aviation Regulation of the Islamic Republic of Iran for the time being in force and subject to the conditions specified below, the Civil Aviation Authority of the Islamic Republic of Iran (CAAIRI) hereby certifies:

Iran Aircraft Industries Company (IACI)
P.O. Box: 14155-1449, Shahid Lashgeri Highway, After Ekhtatan Bridge, Tehran, Iran

as a maintenance organization in compliance with CAAIRI Part-145, approved to maintain the products, parts and appliances listed in the attached "approval schedule" and issue related certificates of release to service using the above reference.

Conditions:

1. This approval is limited to that specified in the scope of work section of the approved Maintenance Organization Exposition (MOE) as referred to in section A of CAAIRI Part-145, and
2. This approval requires compliance with the procedures specified in the approved MOE, and
3. This approval is valid whilst the approved maintenance organization remains in compliance with CAAIRI Part-145.
4. Subject to compliance with the foregoing condition, this approval shall remain valid until any specified date of expiry, unless the approval has previously been surrendered, suspended or revoked.

Date of original issue: 02 Mar. 2014
Date of this issue: 15 Aug. 2025, Revision No.:00
Date of expiry: 14 Aug. 2026

Signature


 Majid Akhlaghi
 Vice President for Flight Standards



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In the name of God


 Islamic Republic of Iran
 Civil Aviation Authority

Date: 8 October, 2025
Number: 24120431008

Maintenance Organization Approval Schedule

Reference: IR.145.19
Organization name: Iran Aircraft Industries Company (IACI)

CLASS	RATING	LIMITATION	BASE	LINE
Aircraft	A1 Aeroplane above 5700 kg	Bae 146-AVRO146-RJ (Honeywell ALF-500 Series)	☒	-
		Boeing 737-100/400-500(CFM56)	☒	-
		Boeing 757-200/500 (PW 2000)	☒	-
		MD-80 Series (PW JT4D)	☒	-
		Embraer EMB-135/145 (ERJ Corp. AE3007A)	☒	-
		Airbus A310 (GE CF6)	☒	-
CLASS	RATING	LIMITATION		
Engine	B1 Turbine	PW JT3D	Limited as per scope of work in MOE and capability list	
		PW JT3D		
		PW C FT6		
		PW 127		
		GE CF700		
		Honeywell T53		
		RR Corp 250		
		TAY 650		
		JT4D		
		CFM56-3A3		
B3 APU		GTCP-85 Series	Limited as per scope of work in MOE and capability list	

Date of original issue: 02 Mar. 2014
Date of this issue: 15 Aug. 2025, Revision No.:00
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Signature


 Majid Akhlaghi
 Vice President for Flight Standards



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In the name of God


 Islamic Republic of Iran
 Civil Aviation Authority

Date: 8 October, 2025
Number: 24120431008

Maintenance Organization Approval Schedule

Reference: IR.145.19
Organization name: Iran Aircraft Industries Company (IACI)

CLASS	RATING	✓	LIMITATION	
Components (other than complete Engines or APU's)	C3 Electrical Power	☒	As per scope of work in MOE and capability list	
	C4 Equipment	☒		
	C5 Engine - APU	☒		
	C8 Flight Control	☒		
	C9 Fuel	☒		
	C12 Hydraulic Power	☒		
	C13 Instruments	☒		
	C14 Landing Gear	☒		
	C15 Oxygen	☒		
	C16 Propellers	☒		
	C17 Pneumatic & Vacuum	☒		
	C20 Structures	☒		
CLASS	RATING			
Specialised Services	D1 Non Destructive Testing	Eddy Current Testing Liquid Penetrant Testing Magnetic Particle Testing Ultrasonic Testing Radiographic Testing (Only X-Ray)		

These terms approval are limited to those products, parts and appliances and to the activities specified in the scope of work section of the approved Maintenance Organization Exposition (MOE, Part 1.9).
Maintenance Organization Exposition reference: IR.MOE.19

Date of original issue: 02 Mar. 2014
Date of this issue: 15 Aug. 2025, Revision No.:00
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Signature


 Majid Akhlaghi
 Vice President for Flight Standards



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Figure 3-1: Test cell MOE

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3.2 Test Specimen Conformity

This statement confirms that the Low-Pressure Turbine (LPT) Blades for the GT-25 project, specifically the Stage 3 (Part Number 798403) and Stage 4 (Part Number 822504) blades, have been designed, manufactured, processed, and assembled in full conformity with the approved specifications.

All relevant manufacturing processes, materials, construction details, and assembly procedures adhere to the requirements outlined in the referenced documentation, including:

- Prototyping Acceptance Test Forms (AEC_232903, AEC_232904);
- Detailed Drawings (drw-IACO-00016752, drw-IACO-00016731);
- Inspection and Quality Test Plans (AEC_500448_4A, AEC_204846_1A);
- Various material and performance testing protocols (e.g., Creep, Fatigue, Hardness, Microstructure, Natural Frequency, Chemical Composition, Stress Rupture, Tensile Tests: AEC_228479 through AEC_227997);
- Engine Operation and Performance Approval Test Plan (AEC_224467);
- Engine Test Safety Assessment (AEC_229058).

Furthermore, it is confirmed that there are no deviations in key characteristics or non-conformances in the test specimens. The final book codes are AEC_909039 & AEC_909040.

3.3 Pre-Test Inspection Check

After engine assemble, torque inspection check of LPT rotor 4th stage blades (AD: 2021-22-19 and Service Bulletin No., JT8D A6224 revision No.7 date: August 26, 2019) was performed.

3.4 Specials to Test Equipment

–

3.5 Instrumentation/Measurement System Calibration

This section identifies the instrumentation used during the JT8D engine test and summarizes the calibration or verification status of the associated measurement systems at the time of test. The instrumentation addressed in this report is divided into test cell instrumentation and engine-mounted instrumentation.

3.5.1 Classification of Instrumentation

For the purpose of this test, the instrumentation is classified in two categories:

1. Test cell instrumentation

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2. Engine-mounted instrumentation

Test cell instrumentation, including the facility measurement and data acquisition systems, is controlled under the laboratory quality system in accordance with ISO-17025. Calibration status, traceability, and periodic verification of these instruments are maintained by the test facility and are referenced accordingly in the applicable calibration records.

For the engine-mounted instrumentation applicable to this test, the Exhaust Gas Temperature (EGT) measurement system was considered. The system consists of eight EGT thermocouples installed on the engine. Accuracy of the EGT measurement system was checked in accordance with the applicable Component Maintenance Manual (CMM) requirements and found acceptable for test use.

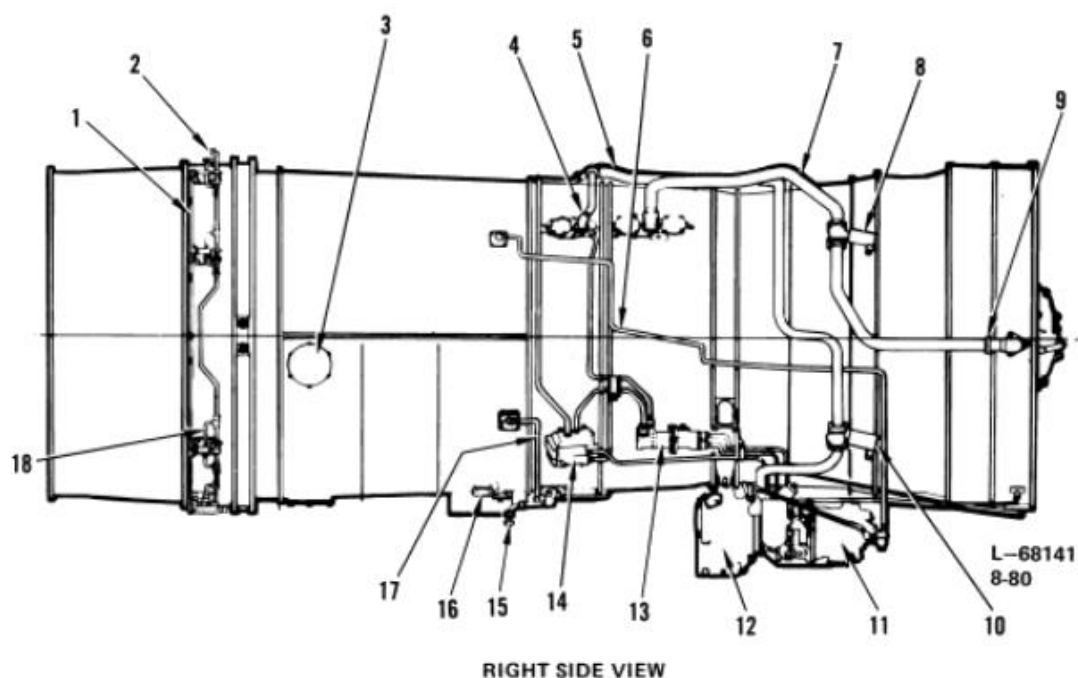
Figure 3-2 and Figure 3-3 shows the location of each sensor and the corresponding instrumentation arrangement. Detailed instrumentation and sensor specification listed in Table 3-1

Table 3-1: Detailed Instrumentation and Sensor Specification Table

	Parameter Name	Sensor	Description	Accuracy
1	CELL TEMP	T AMB	Ambient temperature	± 1 °F (0.55 C°)
2	BAROMETRIC PRESS.	Bar press	Barometric Pressure	± 0.005 PSI (0.034 KPA)
3	COMP. INLET TEMP	TT2	Compressor Inlet Total Temperature	± 2 °F (1.111 C°)
4	PT2	PT2	Compressor Inlet Total Pressure	± 0.01 PSI (0.1 KPA)
5	PS2	PS2	Compressor Inlet Static Pressure	± 0.02 PSI (0.1 KPA)
6	OBS RPM N1	N1	Low Pressure Compressor Rotor Speed	± 0.1%
7	OBS RPM N2	N2	High Pressure Compressor Rotor Speed	± 0.1%
8	TT3	TT3	Low Pressure Compressor Discharge Total Temperature	± 3 °F (1.665 C°)
9	TT4	TT4	High Pressure Compressor Discharge Total Temperature	± 5 °F (2.780 C°)
10	PT4	PT4	High Pressure Compressor Discharge Total Pressure	± 0.5 PSI (3.4 KPA)
11	PCP	PCP	Turbine Cooling Air Pressure	± 0.5 PSI (3.4 KPA)
12	PT6	PT6	High Pressure Turbine Discharge Total Pressure	± 0.5 PSI (3.4 KPA)
13	PT7	PT7	Turbine Discharge Total Pressure	± 0.5 PSI (3.4 KPA)
14	TT7	TT7	Turbine Discharge Total Temperature	± 5 °F (2.780 C°)
15	PT3	PT3	Low Pressure Compressor Discharge Total Pressure	± 0.5 PSI (3.4 KPA)
16	PS3	PS3	Low Pressure Compressor Discharge Static Pressure	± 0.5 PSI (3.4 KPA)
17	OBS THRUST	FN	THRUST	± 46 LB (20.87 Kg)

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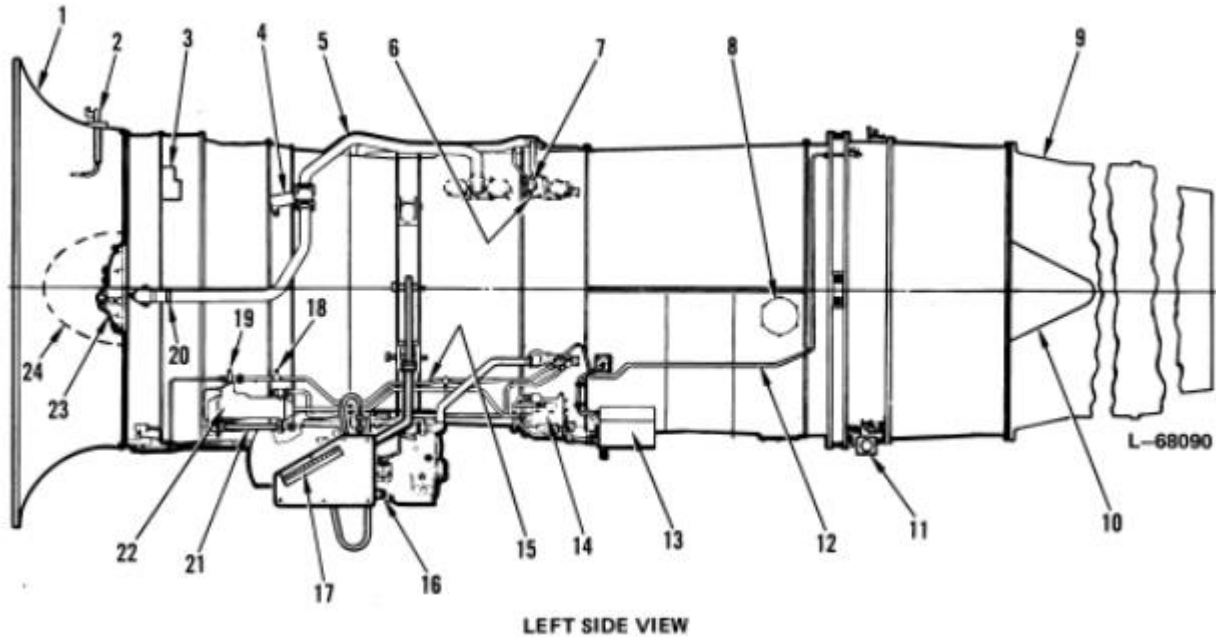
	Parameter Name	Sensor	Description	Accuracy
18	OBS FUEL FLOW	WF	FUEL FLOW	± 1.0% Reading below 4000 PPH and ± 0.75% above 4000 PPH
19	PS4	PS4	High Pressure Compressor Discharge Static Pressure	-
20	BREATHER PRESS	Breather Pressure	Breather Pressure	± 0 - 5 PSI (0 – 34.5 KPA)
21	MAINOIL PRESS	-	-	-
22	FUEL INLET PRESS	-	-	-
23	OIL INLET TEMP	-	-	-
24	OIL OUTLET TEMP	-	-	-
25	FUEL INLET TEMP	-	-	-
26	EXH. STATIC TEMP	-	-	-
27	VIB. NO.1	VIB. NO.1	Inlet (Front) Vibration Pickup	
28	VIB. NO.2	VIB. NO.2	Exhaust (Rear) Vibration Pickup	



- | | |
|---|-------------------------------------|
| 1. Thermocouple (TT7) Harness | 10. Fuel De-Icing Air Shutoff Valve |
| 2. PT7 Connection | 11. Fuel Control |
| 3. Air Bleed Pad (Fan Bypass) | 12. Main Accessory Gearbox |
| 4. Compressor Bleed Valve Supply Connection (PS4) | 13. Bleed Control Valve |
| 5. Fuel De-Icing Air Manifold | 14. Pressure Ratio Bleed Control |
| 6. Fuel Control Burner Pressure Sensing Tube | 15. No. 4 Bearing Seal Drain Tube |
| 7. Anti-Icing Air Tube | 16. Air Bleed Pad (Fan Bypass) |
| 8. Anti-Icing Air Shutoff Valve | 17. Igniter Plug High Tension Lead |
| 9. PWA 45491 Anti-Icing Air Tube Thermocouple | |

Figure 3-2- JT8D Engine Sensor Locations

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- | | |
|---|---|
| 1. PWA 45325 Inlet Bellmouth | 13. Ignition Exciter |
| 2. PWA 45256 Inlet Pressure Probe | 14. Fuel Pressurizing And Dump Valve |
| 3. Front Vibration Pickup (Flange "B") | 15. Main Bearing Oil Pressure Manifold |
| 4. Anti-Icing Air Shutoff Valve | 16. PWA 45507 Oil Out Thermocouple |
| 5. Anti-Icing Air Tube | 17. PWA 46164 (Formerly PWA 22138) Oil Level Sight Gage |
| 6. Low Compressor Air Connection (PS3) | 18. Main Oil Pressure Connection |
| 7. Turbine Cooling Air Connection (PCP) | 19. PWA 45507 Oil In Thermocouple |
| 8. Air Bleed Pad (Fan Bypass) | 20. PWA 45491 Anti-Icing Air Tube Thermocouple |
| 9. PWA 46701 Turbine Exhaust Nozzle | 21. PWA 32935 Dummy Fuel Flowmeter Adapter |
| 10. Tail Cone | 22. Fuel/Oil Cooler |
| 11. Rear Vibration Pickup | 23. PT2 Connection |
| 12. No. 6 Bearing Oil Pressure Tube | 24. PWA 45254 Inlet Cone |

Figure 3-3- JT8D Engine Sensor Locations

3.5.2 Measurement Uncertainty

The methodology for evaluating engine performance uncertainty follows the JCGM 100:2008 (GUM) framework, ensuring a 95% confidence level ($k=2$) across key parameters like Corrected Rotor Speed, EGT_c , EPR, and Specific Fuel Consumption (SFC_c). By assuming a rectangular distribution for sensor error bounds, the analysis converts raw instrument accuracy into standardized uncertainty. This process maintains consistency with engine test-cell reporting and provides an accurate statistical basis for validating thrust and thermodynamic efficiency.

In practice, the calculations are divided into specific regimes, such as SFC_c measurement where uncertainty shifts between 0.75% and 1.0% depending on whether fuel flow is above or below 4000 pph. To ensure diagnostic precision, all temperatures are converted to absolute scales (Rankine) and

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relative uncertainties are maintained as dimensionless ratios within doc. AEC_233071_0A. This systematic approach enables precise engine and performance analysis, ensuring that all derived data remains within the strict limits required for aviation safety and reliability.

4 Test Witnessing

The test was conducted under the supervision of representatives from MAPNA in SAHA aero engine Test-cell. The overall expected values for the engine are presented in the Table 4-1. These values will vary during testing in accordance with the environmental conditions.

Table 4-1: Engine Check Chart (1)

Power Setting	Corrected Jet Thrust LBS (kg)	N1	N2	Corrected TSFC LBS/HR/LB (kg/HR/Kg)
MAXIMUM TAKE-OFF	21,700 (9,841)	7,860	11,760	–
NORMAL TAKE-OFF	21,000 (9,524)	7,670	11,680	0.515 (0.515)
Maximum Climb	18,900 (8,573)	7,240	11,440	0.489 (0.489)
Max. Cruise	16,400 (7,439)	6,890	11,170	0.481 (0.481)

4.1 Test Sequence - B-1

4.1.1 B-1 Preliminary Test and checks, First Day: Thursday, July 02, 2026

Following the assembly of the 4th-stage turbine blades and the completion of the final engine build-up at the workshop, the engine was prepared for preliminary testing. The initial test campaign was successfully conducted as outlined below:

- A thorough visual inspection of the engine and the test stand was carried out.
- The motoring (cranking) test was performed to verify rotor freedom of rotation and to detect any abnormal mechanical noises.
- Engine start-up was executed, and the start-sequence parameters were monitored and validated.
- The engine was tested at idle power, during which functional performance parameters were assessed and leakage checks for air, fuel, and oil systems were conducted.
- Engine shutdown was performed, followed by the Engine Run-Down Time (RDT) test and an assessment for unusual engine noises.
- Visual inspection of the engine and the 4th-stage turbine blades revealed no abnormal conditions.
- Engine start and warm-up at idle power were carried out for a duration of 5 minutes.
- The throttle was advanced slowly and cautiously to the Part Power setting, accompanied by rigorous monitoring of vibration parameters, exhaust gas temperatures, and N1/N2 spool speeds. Initial functional checks of the PRBC (Pressure Ratio Recovery Bleed Control) system and bleed valve closure were also performed.

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- The engine was tested at Part Power for 5 minutes, with comprehensive recording of relevant parameters including thrust, EPR, and others.
- Engine cool-down was conducted at idle power for 5 minutes, followed by the normal engine shutdown procedure.
- The N1 and N2 run-down time test was executed.
- Post-test visual inspections of the engine inlet and exhaust areas, as well as the 4th-stage turbine blades, confirmed no abnormal findings.
- It was scheduled to perform the Notch Torque Check inspection after the engine had completely cooled down.

4.1.2 B-1 Preliminary Tests, Second Day: Date: Wednesday, July 08, 2026

Following the review and successful validation of the previous test results at Part Power, and subsequent to the Notch Torque Check and borescope inspections, the engine test campaign was continued up to the Normal Takeoff power setting. The detailed procedures were as follows:

- The 4th-stage turbine blades Notch Torque Check was performed to verify the structural integrity and proper seating of the 4th-stage blades.
- A borescope inspection of the 4th-stage turbine blades was conducted, and video footage of the blade condition was captured and archived.
- A comprehensive pre-start visual inspection of the engine and the test stand was accomplished.
- Engine start-up was executed, and all start-sequence parameters were closely monitored and verified.
- The engine was warmed up and tested at idle power, including thorough checks of all associated performance parameters and leak tests.
- The throttle was advanced gradually and with extreme caution to the Normal Takeoff power setting. This process involved simultaneous precise monitoring of vibration parameters, temperatures, N1 and N2 spool speeds, and PRBC system operation. Vibration data were continuously monitored and recorded using a vibration analysis system. The engine was gradually conditioned within the power range between Part Power and Maximum Cruise Power.
- The engine was tested at the Normal Takeoff power setting for a duration of 4 minutes, during which all relevant parameters were accepted, evaluated, and systematically recorded.
- Engine cool-down was performed at idle power, followed by the engine shutdown sequence.
- The N1 and N2 run-down time test was carried out.
- Post-test visual inspections of the engine and the 4th-stage turbine blades were performed; no abnormal conditions or discrepancies were observed.
- The 4th-stage turbine blades Notch Torque Check was conducted once again.
- A borescope inspection of the 4th-stage turbine blades was performed, and video documentation of the blade condition was obtained.

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4.1.3 B-1 Functional and Performance Test: July 9, 2026

Following the preparation and execution of the preliminary tests, and after review and evaluation of the test results which were successfully accomplished on the preceding days, the full engine test, “Full Test Run,” was carried out as Test Plan and detailed below.

A) Functional Test

1. A visual inspection of the engine and the test stand was conducted prior to start.
2. The engine was started, and the start sequence parameters were monitored and verified.
3. The engine was warmed up and tested at Idle, with all operational performance parameters reviewed and air, fuel, and oil leakage checks performed.
4. The throttle was advanced to near Normal Takeoff Power and then retarded to Idle to test the Bleed Valve system performance, with associated parameters recorded.
5. The Approach Idle setting was selected via the selector switch, and the engine was tested at Flight Idle (F.I.), with various operational parameters monitored and evaluated.
6. A vibration survey was conducted from Idle to Maximum Takeoff, and the test results were recorded. Simultaneously, data from the vibration analyzer was reviewed and documented.
7. The engine was tested at Part Power, and associated parameters were reviewed and evaluated.
8. The engine was tested at Normal Takeoff, with associated parameters reviewed, evaluated, and recorded. Anti-icing and Fuel Heater systems were functionally tested.
9. The engine was tested at Maximum Takeoff Power, and the ART system was functionally tested.
10. A deceleration run was performed, and the Deceleration Bleed Control system functionality was verified.
11. The engine was tested at Approach Idle, and relevant parameters were reviewed, evaluated, and recorded.
12. An acceleration run was performed, and engine acceleration characteristics (including acceleration time and Thrust Response) were reviewed, evaluated, and recorded.
13. The engine was cooled at Idle, followed by the shutdown procedure.
14. The N1 and N2 Run-Down Time test was performed.
15. Visual inspections of the engine inlet, exhaust, and the 4th-stage turbine blades were conducted; no abnormal conditions were observed.

B) Performance Test

1. The EGT Spread Check electrical system was installed on the engine.
2. A visual inspection of the engine and the test stand was conducted prior to start.
3. The engine was started and warmed up at Idle, and the initial EGT Spread Check was carried out. (During this check, a discrepancy was noted in the No. 8 probe indication channel, which was subsequently troubleshooted and rectified through corrective action.)
4. The throttle was set to Normal Takeoff (Band A), stabilized for 3 minutes, and all test parameters were recorded.
5. The throttle was adjusted to Maximum Continuous Power (Band B), stabilized for 3 minutes, and all test parameters were recorded.

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6. The throttle was adjusted to Maximum Cruise Power (Band C), stabilized for 3 minutes, and all test parameters were recorded.
7. The throttle was adjusted to the Calibration Point, stabilized for 3 minutes, and all test parameters were recorded.
8. The throttle was adjusted to Part Power, stabilized for 3 minutes, and all test parameters were recorded.
9. The throttle was adjusted to Minimum Idle, stabilized for 5 minutes, and all test parameters were recorded.
10. The throttle was advanced to Part Power to re-verify the throttle angle at this setting.
11. The throttle was adjusted to Flight Idle, stabilized for 5 minutes, and all test parameters were recorded.
12. The engine was cooled at Idle, followed by the shutdown procedure.
13. The N1 and N2 Run-Down Time test was performed and recorded.
14. After completion of the test, a final visual inspection of the engine, with particular emphasis on the 4th-stage turbine blades, was conducted; no abnormal conditions were observed.



Figure 4-1: ENG. Starting Shut Down during B-1 Test Runs

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4.2 Test Sequence - B-2

Following the successful completion of phase B-1 test runs, during which new LPT blades stage 4 were tested on the engine (S/N:725746), the blades were subsequently disassembled and subjected to required inspections, including visual test and Fluorescent Penetrant Inspection (FPI). The inspection results indicated that all blades were in ideal condition.

Accordingly, LPT stage 3 and stage 4 blades, after undergoing assembly, static balancing, and dynamic balancing, were installed on the engine. The engine was then prepared for Phase B-2 testing, and the initial test was successfully conducted as detailed below

4.2.1 B-2 Preliminary Test and checks, First Day: Saturday, 17 Mordad 1405 (August 8, 2026)

1. Visual inspection of the engine and test cell was performed.
2. Motoring test was carried out, with focus on the LPT stage 3 and 4 blade areas to check for abnormal noises and rotor rotational smoothness.
3. Engine start and ignition were executed, and start process parameters were monitored.
4. Engine test at idle speed for 5 minutes, with performance parameter monitoring and leakage checks (air, fuel, oil).
5. Engine test at approach idle speed, with performance parameter monitoring and verification.
6. Return to idle speed, normal engine shutdown, and RDT¹ and EUN tests were performed.
7. After shutdown, visual inspection of the exhaust, engine and LPT stage 4 blades was conducted.
8. Engine restart, ignition, and warm-up at idle speed for 5 minutes.
9. Slow and cautious throttle advancement to part power, with close monitoring of vibration, temperature, N1 and N2 speeds, and preliminary check of PRBC system operation.
10. Engine run at part power for 5 minutes, with performance parameter monitoring.
11. Return to idle speed, engine cooling for 5 minutes, and normal shutdown.
12. RDT and EUN tests performed.
13. Visual inspection of engine inlet and exhaust, and LPT stage 4 blades – no abnormal conditions observed.
14. Engine restart, ignition, and warm-up at idle speed.
15. Slow and cautious throttle advancement to normal takeoff power, with close monitoring of vibration, temperature, N1, N2, PRBC system operation, and vibration parameters recorded via vibration analyzer; engine warm-up within the range between part power and maximum cruise power.
16. Engine test at normal takeoff power for 4 minutes, with parameter recording and verification.
17. Slow throttle retraction to idle speed with continuous monitoring of vibration, temperature, N1, N2, PRBC system, and vibration data recording via analyzer.
18. Engine cooling at idle speed for 5 minutes, normal shutdown, and RDT and EUN tests.

¹ Run Down Time

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19. After engine cool-down, notch torque check and borescope inspection were performed – no abnormalities observed.

4.2.2 B-2 Functional and Performance: Sunday, 18 Mordad 1405 (August 9, 2026)

Following review and evaluation of the initial test results up to normal takeoff power, and subsequent acceptance of the notch torque and borescope inspection results, the full engine test run was carried out as detailed below:

A) Functional Tests

1. The EGT spread check electrical system was installed on the engine.
2. A visual inspection of the engine and the test cell was performed prior to start.
3. The engine was started, and the start sequence parameters were monitored and verified.
4. The engine was warmed up and tested at Idle, with all relevant parameters reviewed and air, fuel, and oil leakage checks performed.
5. The Bleed Valve system functional test was conducted, and associated parameters were recorded.
6. The engine was warmed up to approximately Part Power.
7. A Vibration Survey was carried out from Idle up to Maximum Takeoff, and the test results were recorded. Simultaneously, data from the vibration analyzer was reviewed and documented.
8. Approach Idle was selected via the selector switch, and the speed was stabilized.
9. An Acceleration Check was performed to evaluate engine acceleration characteristics, including acceleration time and Thrust Response; relevant parameters were reviewed, evaluated, and recorded.
10. Noting a noticeable change in the engine acceleration time compared to the Phase B1 test, and in order to ensure full confidence in the recorded value, the acceleration check was repeated twice to verify the measured time.
11. The engine was tested at Idle, and associated parameters were reviewed, evaluated, and recorded.
12. The engine was tested at Approach Idle, and associated parameters were reviewed, evaluated, and recorded.
13. The engine was tested at near Band C, and associated parameters were reviewed and evaluated.
14. The engine was tested at Normal Takeoff Power, accompanied by a re-check Vibration Survey; associated parameters were reviewed, evaluated, and recorded; the Anti-icing and Fuel Heater systems were functionally tested.
15. The ART system was activated, and the engine was tested at Maximum Takeoff Power with relevant parameters recorded.
16. The speed was stabilized at Idle for 5 minutes, and all associated parameters were reviewed.
17. The speed was stabilized at Approach Idle, and all associated parameters were reviewed.
18. The engine was cooled at Idle and shut down normally; the N1 and N2 Run-Down Time test was performed and recorded.

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19. Visual inspections of the engine inlet, exhaust, and the 4th-stage turbine blades were conducted; no abnormal conditions were observed.
20. Following the review of test results up to this stage and a technical discussion regarding the recorded acceleration time, it was decided to also perform an alternative acceleration test method (in accordance with 14 CFR Part 33.73). The alternative acceleration test is required to be performed as follows:
 - a) Set the throttle to 15% of Takeoff thrust.
 - b) Advance the throttle rapidly (within less than one second) to Takeoff Power.
 - c) Record the Thrust Response time from the initiation of throttle movement until reaching 95% of Rated Takeoff Power.
 - d) The recorded time must not exceed a maximum of 5 seconds.

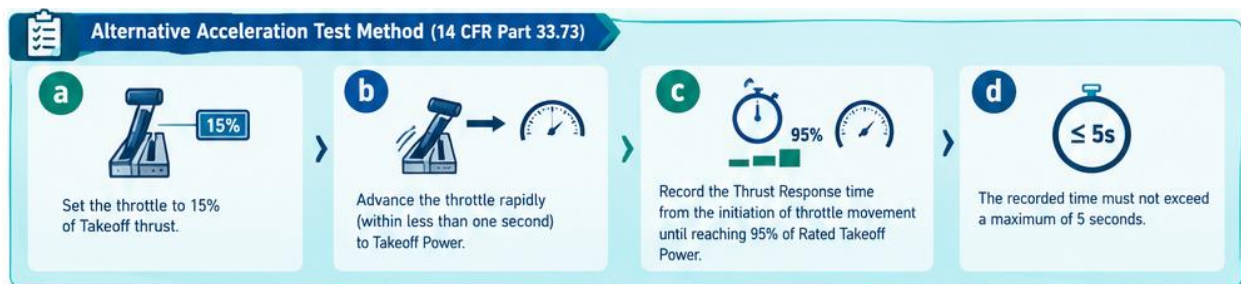


Figure 4-2: Sequence of Acceleration test 33.73

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Figure 4-3: B-2 Functional Tests

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B) Performance Test

1. A visual inspection of the engine and the test stand was performed prior to start.
2. The engine was started, run, and warmed up at Idle for approximately 6 minutes.
3. The throttle was advanced to Normal Takeoff Power, a Vibration Survey was conducted, and the final engine vibration status was reviewed.
4. The throttle was retarded to Idle and stabilized for approximately 2 minutes.
5. The throttle was advanced to Normal Takeoff Power, the ART system was activated (Band A'), and the engine was stabilized for 2 minutes; all test parameters were recorded. The throttle was then returned to Normal Takeoff Power (Band A), stabilized for 3 minutes, and all test parameters were recorded.
6. The throttle was set to Maximum Continuous Power (Band B), stabilized for approximately 3 minutes, and all test parameters were recorded.
7. The throttle was set to Part Power, stabilized for approximately 3 minutes, and all test parameters were recorded.
8. The throttle was set to Maximum Cruise Power (Band C), stabilized for approximately 5 minutes, and all test parameters were recorded.
9. The throttle was set to Minimum Idle, stabilized for approximately 3 minutes, and all test parameters were recorded.
10. Approach Idle was selected, stabilized for 3 minutes, and all test parameters were recorded.
11. The throttle was returned to Idle and stabilized for 7 minutes; the parameters recorded throughout the Performance test were reviewed.
12. The engine was stabilized at 15% of NTO (Normal Takeoff) thrust, and the alternative acceleration test method was performed, with the Thrust Response time recorded.
13. The throttle was returned to Idle, the engine was cooled for 5 minutes, and normal shutdown was performed.
14. The N1 and N2 Run-Down Time test was performed and recorded.
15. Upon test completion, a final visual inspection of the engine, with particular emphasis on the 4th-stage turbine blades, was conducted; no abnormal conditions were observed.
16. After the engine had cooled, the Notch Torque Check and Borescope Check were performed; no abnormal conditions were observed.

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Figure 4-4: B-2 Performance Test Sequence

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4.2.3 B-2 Repeatability and Ambient Condition Check, Date: Monday, 19 Mordad 1405 (August 10, 2026)

To repeat the tests and further investigate results under different ambient temperatures, the following test runs were conducted in the presence of representatives from the CAA.IRI and senior officials from SAHA and MAPNA Aero. Throughout the test period, all technical inquiries from the authority representatives regarding the engine test, various parameters, and operating envelopes were addressed with full technical explanations.

1. Visual inspection of engine and test stand prior to start.
2. Engine start and ignition, with start process parameter monitoring.
3. Engine warm-up and test at idle, with all relevant parameter checks and leakage tests (air, fuel, oil).
4. Bleed valve system functional test and parameter recording.
5. Engine warm-up at maximum cruise power.
6. Vibration survey from idle to maximum takeoff power, with result recording and simultaneous verification using vibration analyzer.
7. Engine test at part power, with parameter review and evaluation.
8. Engine test at normal takeoff power, with parameter review, evaluation, and recording.
9. Engine cooling at idle, normal shutdown, and RDT and EUN tests.
10. In the presence of the authority representatives, visual inspection of engine inlet, exhaust, and LPT stage 4 blades was performed – no abnormal conditions observed.
11. To fulfill another acceleration test requirement (per 14 CFR Part 33.73) from idle to NTO, the engine was restarted and the test was successfully conducted as follows:
12. Visual inspection of engine and test stand prior to start.
13. Engine start and warm-up at idle for 5 minutes.
14. Engine warm-up at part power for 3 minutes.
15. Engine test at normal takeoff power, with parameter review, evaluation, and recording.
16. Stabilization at idle for 3 minutes.
17. Acceleration test performed (throttle advancement from idle to NTO in under 1 second), with monitoring of relevant parameters including surge and stall conditions during acceleration.
18. Engine cooling at idle, normal shutdown, and RDT and EUN tests.
19. Visual inspection of engine inlet, exhaust, and LPT stage 4 blades – no abnormal conditions observed.
20. It was decided that after the engine is returned to the workshop, a notch torque check will be performed, and following disassembly of the engine and LPT, required inspections including VT and FPI on LPT stage 3 and stage 4 blades will be conducted per the established plan.

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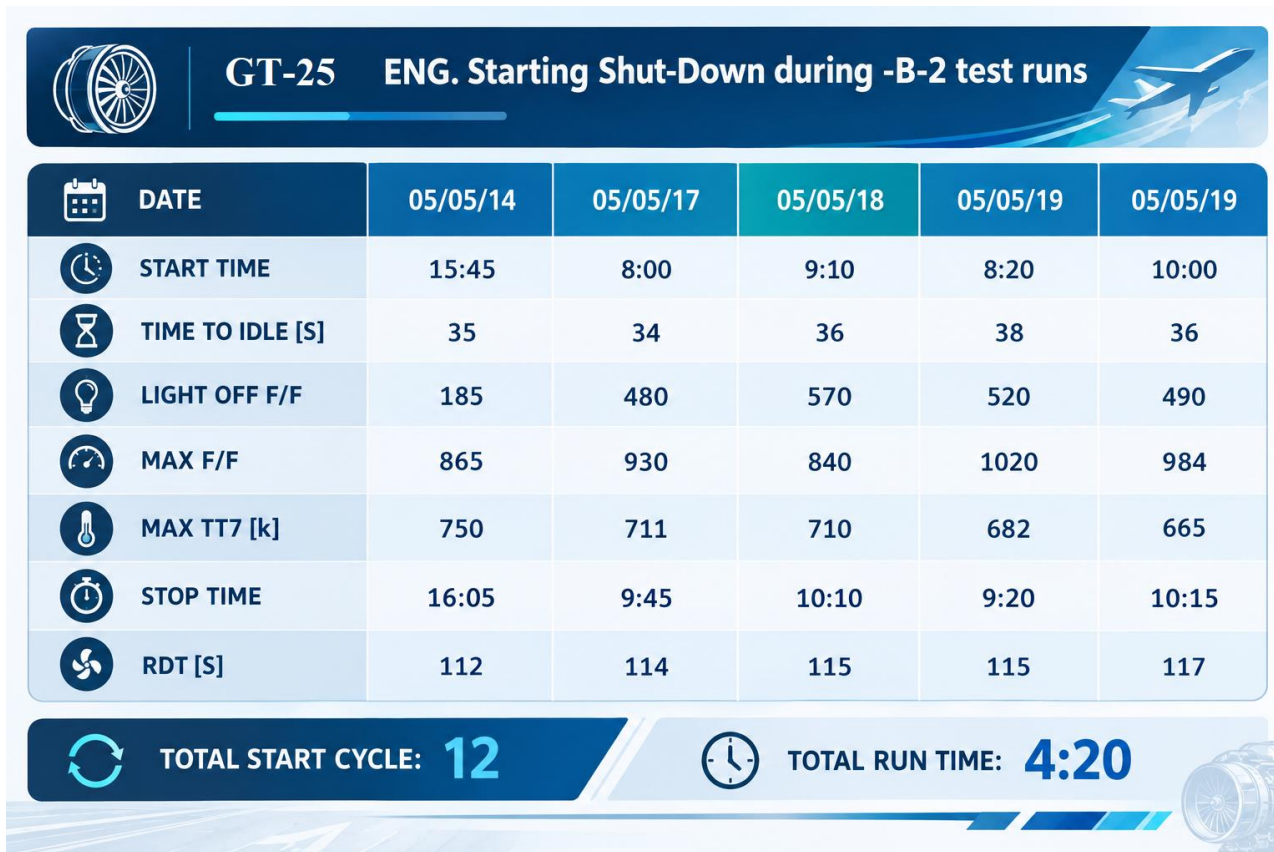


Figure 4-5: ENG. Starting Shut-Down during -B-2 test runs

4.3 On-Test Inspections

The on-test inspections performed during this test are detailed below.

4.3.1 Notch torque and Borescope Checks

Notch Torque and borescope inspections were performed in accordance with the test plan (2) after each phase in the test cell. In total, no anomalies were observed during the tests.

4.4 Test Deviations

Overall, the Phase B-2 test was carried out in full compliance with the engine technical manual and the test schedule, and was successfully completed within a duration of 7:20 hours. In all engine operating regimes, the performance parameters remained within the permissible limits, with the exception of the engine vibration at the fan inlet case. During steady normal take-off, in Test B-1, the inlet vibration exceeded the acceptance limit, reaching 3.2 mils. However, in Test B-2, the inlet vibration improved, while the exhaust vibration exceeded the limit, recording values between 3.2 and 3.6 mils.

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The measured tip clearance exceeds the allowable limit specified in the manual. This condition is attributed to the engine's OEM honeycomb seals. Such an increase in clearance will subsequently lead to a degradation in performance.

5 Conclusion

The B-1 and B-2 test campaigns were completed in accordance with the approved test plan and applicable engine technical manual. Test-cell instrumentation and data acquisition were controlled under the laboratory quality system in accordance with ISO 17025, and measurement uncertainty was evaluated using the JCGM 100:2008 (GUM) framework at a 95% confidence level. Functional and performance testing, including EGT spread checks, acceleration and Thrust Response evaluations, vibration surveys, RDT/EUN tests, and notch torque and borescope inspections, confirmed stable engine operation. Phase B-2 was completed in totally 7:20 hours, and all performance parameters remained within permissible limits in all operating regimes, with the exception of vibration. No abnormal conditions were observed during the post-test inspections.

The primary deviation was vibration: in B-1, fan inlet vibration exceeded the acceptance limit, reaching 3.2 mils during steady normal takeoff; in B-2, inlet vibration improved, but exhaust vibration exceeded the engine acceptance limits, recording values between 3.2 and 3.6 mils. In addition, the measured tip clearance exceeded the allowable limit specified in the manual, attributed to the engine's OEM honeycomb seals, which is expected to lead to performance degradation. These findings do not invalidate the overall test results, but they require engineering disposition and corrective action for OEM engine acceptance.

The test results documented herein demonstrate full compliance with the requirements of 14 CFR Part 33, Sections 33.73 and 33.89 (a) (1-4) and (b). All acceptance criteria defined in the approved Test Plan were met.

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6 Definitions/Abbreviations/Acronyms

Abbreviation / Symbol	Definition / Expansion	Unit
ART	Automatic reverse thrust	
CFR	Code of Federal Regulations	
CMM	Component Maintenance Manual	
EGT	Exhaust Gas Temperature	
EGT_C	Corrected Exhaust Gas Temperature	
EPR	Engine Pressure Ratio (PT7/PT2)	
EUN	Engine Unusual Noise	
F.I.	Flight Idle	
F/N	Thrust	LBS
F.P.I.	Fluorescent Penetrant Inspection	
GUM	Guide to the Expression of Uncertainty in Measurement	
ISO 17025	ISO/IEC 17025	
LPT	Low Pressure Turbine	
N1	Low Pressure Compressor Rotor Speed	R.P.M
N1 RDT	N1 Run Down Time	sec
N2	High Pressure Compressor Rotor Speed	R.P.M
N2 RDT	N2 Run Down Time	sec
NTO	Normal Takeoff (Band A)	
OBS FUEL FLOW	Observe FUEL FLOW	LBS/HR
OBS RPM N1	Observe RPM N1	R.P.M
OBS RPM N2	Observe RPM N2	R.P.M
OBS THRUST	Observe Thrust	LBS
OEM	Original Equipment Manufacturer	
PBAR	Barometric Pressure	IN.HGA
PCP	Turbine Cooling Air Pressure	IN.HGA
PRBC	Pressure Ratio Bleed Control	
PT 7	Turbine Discharge Total Pressure	IN.HGA
PT2	Compressor Inlet Total Pressure	IN.HGA
RDT	Run-Down Time	
SFC	Specific Fuel Consumption	
TSFC	Thrust Specific Fuel Consumption	
TT 7 (EGT)	Turbine Discharge Total Temperature	°F

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Abbreviation / Symbol	Definition / Expansion	Unit
TT2 (CIT)	Comp. Inlet Temperature	°F
VIB. NO.1	Vibration NO.1	MIL'S
VIB. NO.2	Vibration NO.2	MIL'S
VT	Visual Test / Visual Inspection	

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7 References

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8 Approval

Table 8-1: Approval of Test Witness Report.

Roles	Signature/Date
Prepared by: DE [Iman Laraibi] [Aghil FazelNiari] [Vahid Aramian]	
Provided Proof of Compliance by: DOM [Golsa Ghanati]	
Checked by: CVE [Golsa Ghanati] [Alireza Navaei] [Ali Farokhipour]	
Approved by: HOA [Alireza Taghipour]	